

COORDINATE GEOMETRY: DISTANCE & MIDPOINT FORMULA



WHAT IS THE COORDINATE PLANE?

The coordinate plane is a flat surface made up of two number lines that cross each other at right angles.

The horizontal line is called the X-axis – it runs left and right. The vertical line is called the Y-axis – it runs up and down. Where they meet is called the origin (0, 0). Together, these axes create a grid where we can locate any point using two numbers!

"A playground for points!"

Coordinates are written as (x, y) – an ordered pair of numbers.

x is the horizontal position – move left or right first.

y is the vertical position – move up or down second.

WHAT IS DISTANCE BETWEEN TWO POINTS?

In coordinate geometry, distance tells us how far apart two points are on the coordinate plane.

Imagine two friends standing at different spots in a park – the distance is simply the length of the straight line connecting them!

We can calculate this using a special formula:

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Don't worry, we'll break this down step by step on the next slide!

THE DISTANCE FORMULA

$$\text{Distance} = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$

How it works:

- $(x_2 - x_1)$ = difference between x-coordinates
- $(y_2 - y_1)$ = difference between y-coordinates
- Square each difference to remove negatives
 - Add the squares together
- Take the square root to find the actual distance!

EXAMPLE: FINDING DISTANCE

Let's find the distance between points A(2,3) and B(7,8)!

Step 1: Identify coordinates

$$x_1 = 2, y_1 = 3 \text{ and } x_2 = 7, y_2 = 8$$

Step 2: Apply the formula

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\text{Distance} = \sqrt{(7 - 2)^2 + (8 - 3)^2}$$

$$\text{Distance} = \sqrt{5^2 + 5^2} = \sqrt{25 + 25} = \sqrt{50} \approx 7.07 \text{ units}$$

✨ The distance between (2,3) and (7,8) is $\sqrt{50}$ or approximately 7.07 units!

PRACTICE TIME!

DISTANCE

Now it's your turn to try!

Question: What is the distance between the points (1, 2) and (5, 6)?

Use the distance formula:

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Plot the points on your coordinate plane, then calculate step by step. Write your answer and compare with a classmate!

WHAT IS THE MIDPOINT?

The midpoint is the exact middle point between two points on a coordinate plane.

Think of it as the halfway spot!

To find the midpoint, we simply take the average of the x-coordinates and the average of the y-coordinates of our two points.

If you have two points, the midpoint is where you would meet your friend if you both walked the same distance toward each other. It divides the line segment connecting the two points into two equal parts.

THE MIDPOINT FORMULA

The midpoint formula helps us find the exact center point between two coordinates on a plane.

$$\text{Midpoint} = ((x_1 + x_2)/2, (y_1 + y_2)/2)$$

How it works:

- Add the x-coordinates and divide by 2 → gives the middle x value
- Add the y-coordinates and divide by 2 → gives the middle y value
- The result is your midpoint coordinate!

EXAMPLE: FINDING THE MIDPOINT

Let's find the midpoint between points A(4, 6) and B(10, 2).

Step 1: Write the midpoint formula

$$\text{Midpoint} = ((x_1 + x_2)/2, (y_1 + y_2)/2)$$

Step 2: Substitute the values

$$\text{Midpoint} = ((4 + 10)/2, (6 + 2)/2)$$

Step 3: Calculate

$$\text{Midpoint} = (14/2, 8/2) = (7, 4)$$

The midpoint is at (7, 4)! 🎯

PRACTICE TIME!

MIDPOINT

Now it's your turn to practice!

Find the midpoint between the points (3,5) and (7,9).

Remember the formula:

$$\text{Midpoint} = ((x_1 + x_2)/2, (y_1 + y_2)/2)$$

Take a moment to calculate, then check your answer with your partner or teacher!

DISTANCE VS MIDPOINT: QUICK COMPARISON

Understanding the difference between distance and midpoint helps you solve coordinate geometry problems correctly.

Both use the same two points but give you completely different information about them.



Distance measures length:
tells you how far apart two points are on the coordinate plane.



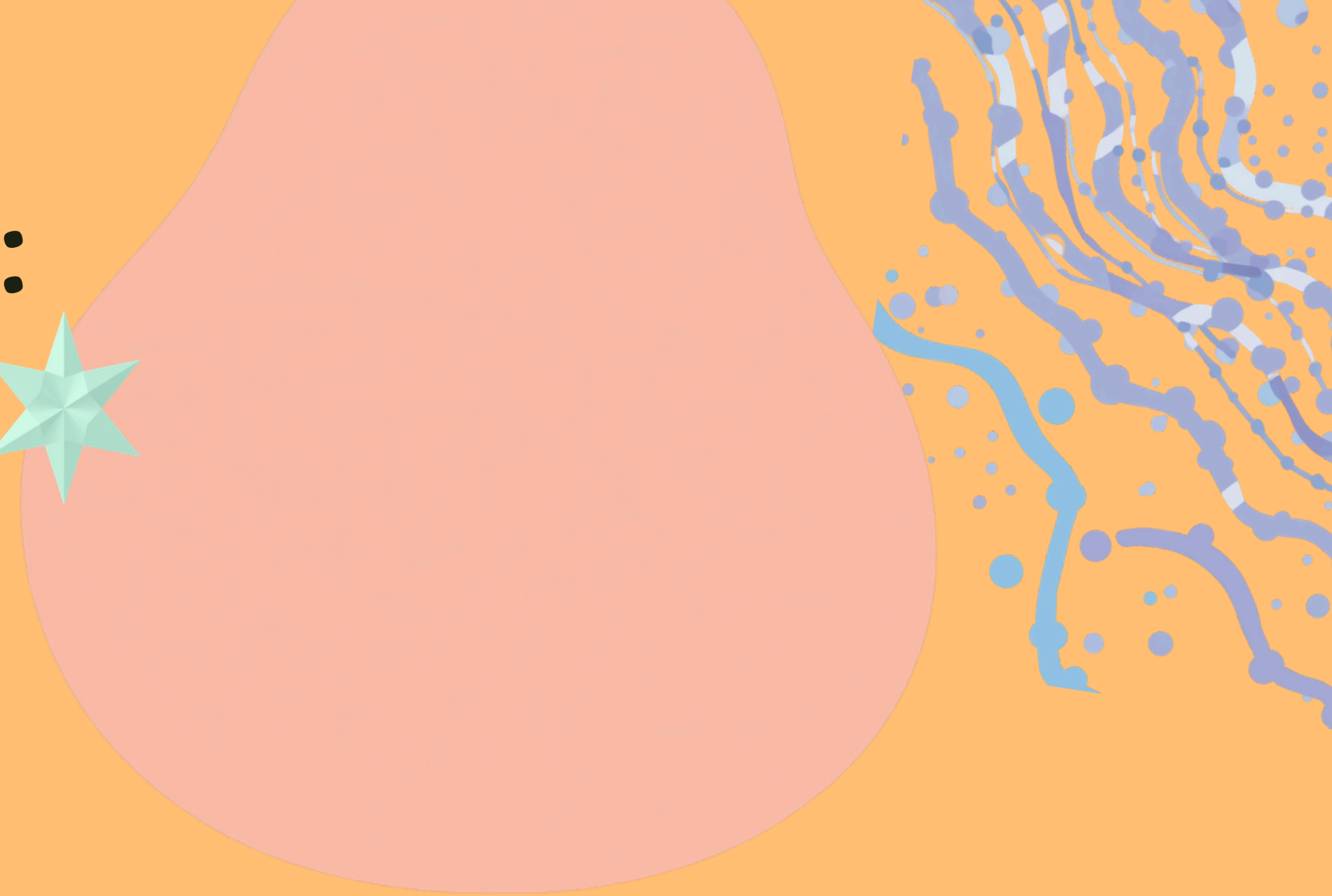
Midpoint finds center:
gives you the exact center point between two points.



Distance formula uses:
uses subtraction, squaring, and square root in its formula.



Midpoint formula uses:
uses addition and division by 2 to find averages.



REAL-LIFE USES OF DISTANCE & MIDPOINT



Distance and midpoint formulas aren't just for math class – they help us solve real problems every day! From planning trips to finding meeting spots, these tools are everywhere.

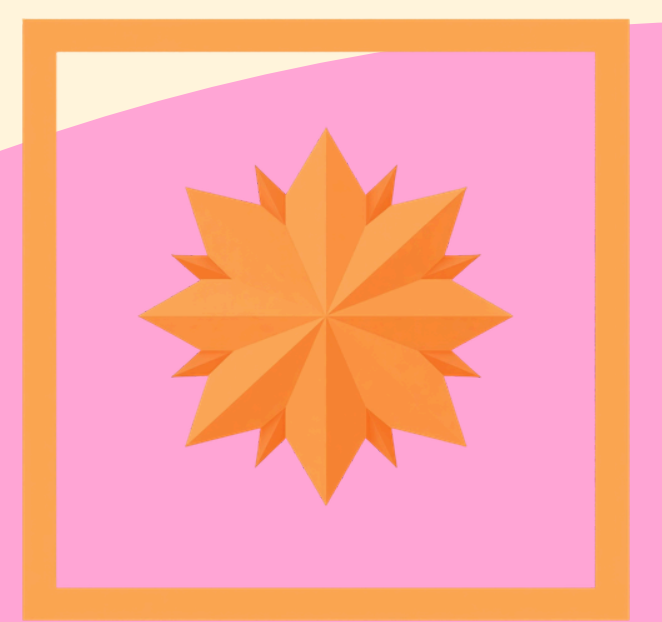
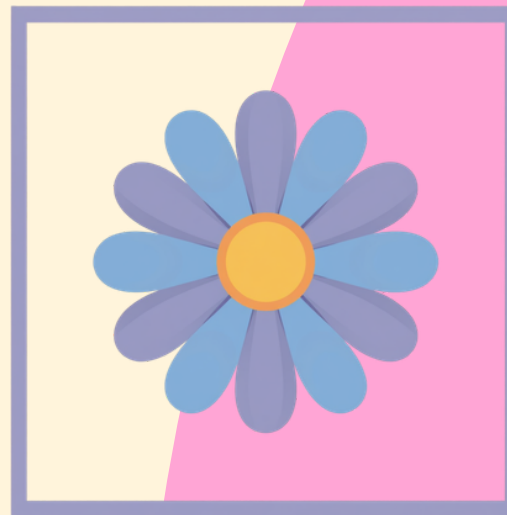
Distance: Measure how far apart two places are on a map or GPS.

Midpoint: Find the halfway point to meet a friend fairly!

Both help in dividing land, planning parks, or placing landmarks.

COMMON MISTAKES TO AVOID

Learning coordinate geometry is fun, but watch out for these common errors! Being aware of these mistakes will help you solve distance and midpoint problems correctly every time.



Forgetting to square the differences in the distance formula before adding them.

Mixing up x and y values when substituting coordinates into formulas.

Not dividing by 2 in the midpoint formula – remember to find the average!

Making calculation errors in subtraction or addition – always double-check your work!

SUMMARY: WHAT WE LEARNED TODAY

Great job exploring coordinate geometry! Today we discovered how to navigate the coordinate plane and use powerful formulas to measure distances and find midpoints between any two points.



The coordinate plane uses X and Y axes to locate any point as (x, y) .

Distance formula: $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$ measures how far apart points are.

Midpoint formula: $((x_1 + x_2)/2, (y_1 + y_2)/2)$ finds the center between two points.

CHALLENGE QUESTION!

Put your skills to the test! Use both formulas you've learned today.

Find the distance AND midpoint between the points:

(1, 4) and (5, 10)

Try it on your own first, then check with a partner!

Remember: Distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Midpoint = $((x_1 + x_2)/2, (y_1 + y_2)/2)$

KEEP EXPLORING COORDINATES!

Math is the key to unlocking the world around you! Every point you plot, every distance you calculate, and every midpoint you find brings you closer to understanding the amazing patterns in our world. Keep practicing, stay curious, and remember – coordinate geometry is your superpower for solving real-world problems!